Assignment 9 — CNN and Pretrained Vision Transformer

Deep Learning | Practical Report / Solution

BRACT's Vishwakarma Institute of Technology, Pune

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Problem Statement

Perform image classification on the CIFAR-10 dataset using a CNN model trained from scratch and a pretrained Vision Transformer (ViT), then compare their performance using accuracy, precision, recall, F1-score, confusion matrices, training time, and a comparison table.

Objective

Build and evaluate two image-classification approaches on CIFAR-10: a conventional convolutional neural network and a pretrained Vision Transformer fine-tuned for the dataset's ten classes.

Dataset

CIFAR-10 contains 60,000 colour images of size 32×32 across ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The notebook uses 50,000 training images, splits training data into training and validation subsets, and evaluates on the 10,000-image test set.

Methodology

1. CNN from scratch: Images are converted to tensors and normalized using CIFAR-10 channel statistics. A CNN with convolution, batch normalization, ReLU activations and max-pooling layers is trained for five epochs. Training and validation loss/accuracy are tracked.

2. Pretrained ViT: The notebook loads **google/vit-base-patch16-224-in21k**, prepares CIFAR-10 images using the matching Hugging Face image processor, resizes them to the model's expected input, replaces the classification head with ten output labels, and fine-tunes the model.

3. Evaluation: Both models are evaluated on the CIFAR-10 test set. Accuracy, weighted precision, weighted recall, weighted F1-score, classification reports and confusion matrices are generated. Training time and predictions are also compared.

Training Configuration

CNN: 5 epochs. Pretrained ViT: 2 epochs, learning rate 2×10■■. ViT batch size is configured as 16 when CUDA is available and 4 otherwise. Actual training times are taken from the saved notebook outputs.

Recorded Results

Metric	CNN	Pretrained ViT
Accuracy	0.7025 (70.25%)	0.9845 (98.45%)
Precision (weighted)	0.7354	0.9845
Recall (weighted)	0.7025	0.9845
F1-score (weighted)	0.7069	0.9845
Training time	1.53 min	18.61 min

The saved notebook outputs report that the pretrained ViT achieved the higher test accuracy. Its accuracy was 28.20 percentage points above the CNN. The CNN trained much faster in this run.

Conclusion

The experiment compares a CNN trained from scratch with a pretrained Vision Transformer on CIFAR-10. Based on the recorded outputs, the CNN achieved 70.25% test accuracy, while the pretrained ViT achieved 98.45%. The ViT therefore performed better on the reported classification metrics, although it required substantially more training time (18.61 minutes compared with 1.53 minutes for the CNN). The results demonstrate the benefit of transfer learning from large-scale image pretraining for this experiment.

Source-file consistency note

The uploaded notebook is Assignment 9 and implements CIFAR-10 image classification using CNN and ViT. However, the uploaded README (3).md is titled “Assignment 10: Audio Classification with Speech-to-Text (STT) and Text-to-Speech (TTS)” and describes a different audio project. This report and the accompanying README are based on the Assignment 9 notebook; the supplied audio README should not be placed in this Assignment 9 folder.